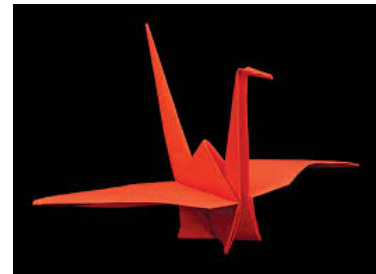
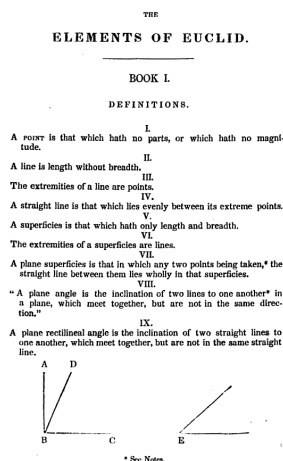


Mathematics in origami

Origami, the art of paper folding, has a long history. In origami, people can make beautiful handicrafts using only one piece of paper. There are various kind of techniques of paper folding that are used in origami and none of them are complicated to learn. But, since the early 20th century, origami has become more than a relaxing hobby. As more and more researches were carried out to study the essence of origami, mathematicians realized that there are some serious maths behind paper folding, which would start a completely new area of geometry.

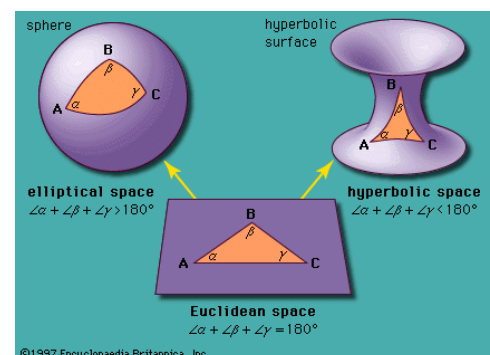


You may be wondering: how can folding a piece of paper leads us to some serious maths? The reason why mathematicians are excited about paper folding geometry is that it has some properties that are not possessed by traditional geometry which uses ruler and compass only, and these properties are used to solve some problems that are proved cannot be solved in traditional geometry, for example, trisecting an angle, which will be discussed later. Why can origami do these amazing things? The implicit answer is hidden in the axiom system of origami geometry, which is completely different from the system of Euclidean geometry. And what are axioms? To understand how axioms work, we must go back to ancient greek.



At the start of his great book "Elements", Euclid wrote 5 axioms - 5 simple and obvious fact that he believed proof is not needed, for instance, he stated that you can draw a straight line using 2 point, which is a fairly obvious fact to everyone. After stating these 5 axioms, Euclid systematically and logically started to construct the whole world of traditional geometry, from some small and simple facts to deep results. This is saying that Euclid took these 5 axioms as a starting point, and proved all the subsequent theorems using ONLY these 5 axioms. In the following 2000 years, Euclid's 5 axioms are regarded as the unbreakable rule of geometry, and considered to be the base of mathematics. From this, we can see how important the concept of axiom is to any logical system - anything within this system comes from the axioms of this system.

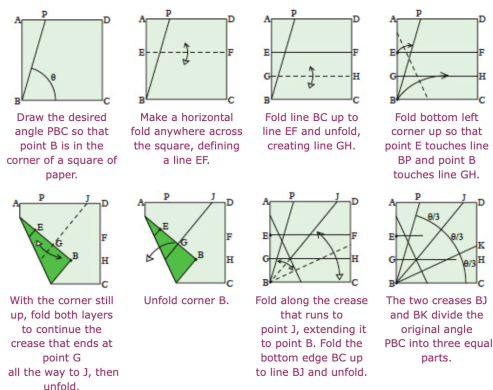
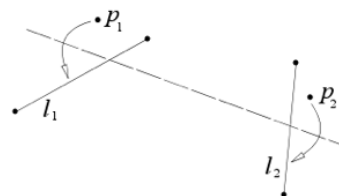
If the axioms of a system is modified, even just by a tiny bit, you will get completely different results. For example, from Euclid's 5 axioms, we get "The sum of angles in a triangle is 180", however, if we change the 5th axiom, famously known as parallel postulate, to something else, then the sum of angles in a triangle won't be 180 anymore. How does this happen? If we take a deeper look on those axioms, we will find that Euclid assumed everything in his geometric world to be plain. If we want to study the geometry that is based on a curved surface instead of a plain, then the parallel



postulate will no longer be true, therefore it must be replaced by an axiom that suits the curved surface, and that is how the “Non-Euclidean geometry” started.

The same thing works for origami geometry. The Euclidean geometry is entirely based on ruler and compass, but the origami geometry is based on folding, so obviously those 5 axioms are not suitable any more, and a new set of axioms should be defined for origami geometry. A set of 7 axioms is discovered to describe the origami geometry. The first complete statement of the seven axioms of origami by French folder and mathematician Jacques Justin was written in 1986, but were overlooked until the first six were rediscovered by Humiaki Huzita in 1989.

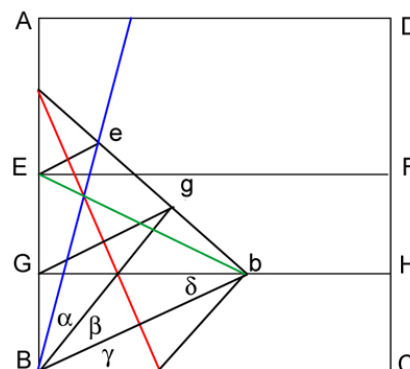
Now let's look these axioms in details. They are fairly simple and they will suit your understanding of paper folding really well. The first axiom states that given two lines l_1 and l_2 , there is a fold that places l_1 onto l_2 . This is exactly the same thing as connecting 2 points by a straight line in traditional geometry. Axiom 2 says that given two points p_1 and p_2 , there is a unique fold that places p_1 onto p_2 . You can try this on a piece of paper and you will find that this axiom actually states that you can construct the perpendicular bisector in origami geometry. Axiom 3, 4 and 5 state that you can find the bisector of an angle, the perpendicular line passing through a given point and intersection of a line with a circle respectively in origami geometry. When we get into axiom 6, it becomes more interesting, and it is the key to trisect an angle in origami geometry. It states that given two points p_1 and p_2 and two lines l_1 and l_2 , there is a fold that places p_1 onto l_1 and p_2 onto l_2 .



How can you trisect an angle in origami? First we make a random angle by folding, (look at the diagram on the left) then fold the paper and get a crease EF . Next thing to do is to fold the bottom to EF so we get a crease GH . Now we get 2 creases. We then mark 2 points: B and E . Here is the most interesting part: apply axiom 6, we fold so that the point B is placed onto the crease GH and E is placed onto the edge of the angle. Now we mark 2 new point on the GH crease: where B is placed is marked B' and the intersection of the GH crease with the crease we just created is marked as I . The segment BB' and BI are the line that trisect the angle.

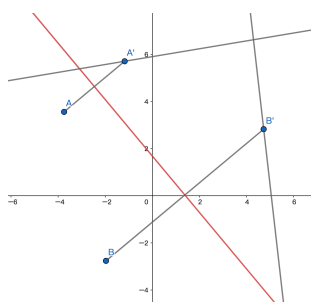
You may be very confused by this method, but it is indeed true and it is not hard to prove. In the image on the right, it is easy to prove that triangle EBb is isosceles. The mirror image of the triangle EBb about the red crease is the triangle eBb , which is also isosceles and congruent with triangle EBb . Therefore we can tell that angle alpha, beta, gamma and delta are all equal, so the angle is trisected.

What is the essence behind all of these? Why can origami do it

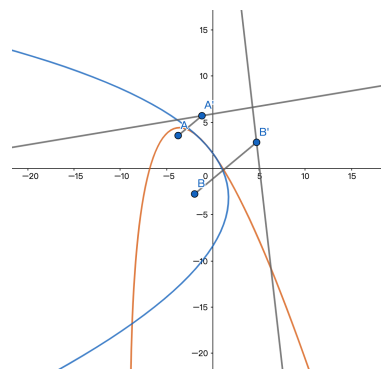


but ruler and compass can't? Trisecting an angle is not the only example. Another famous problem, doubling the cube, which cannot be solved using ruler and compass only, is also solvable in origami geometry. Think about it - to double the volume of a square, you need to construct a segment that has a length of cube root 2. Mathematicians proved doubling a cube is not possible by proving constructing cube roots using ruler and compass only is not possible. Why? Constructing a cube root means finding a root of a cubic equation. We know that the equation for a straight line is linear, the equation for a circle is quadratic. In analytic geometry, using straight line and circle only, you can never get a cubic equation, and that is why constructing a cube root in traditional geometry is impossible. So we can say that anything that involves cube or higher power, is not possible in ruler and compass geometry. Using trigonometry, we can prove that the formula for $\cos(3\theta)$ involves a term that is cubic, therefore the possibility of trisecting an angle is denied in traditional geometry.

$$\cos(3\theta) = 4\cos(\theta)^3 - 3\cos(\theta).$$



You might wonder in which step of the trisecting angle process that cubic is involved. The answer is still axiom 6. The key is to apply the axiom 6 into analytic geometry and see what happens. If we transfer this question to analytic geometry, we get something like this: given 2 points A, B and 2 lines l_1 and l_2 , find a straight line l_3 that the reflection of A about l_3 (A') is placed on l_1 and the reflection of B about l_3 (B') is placed on l_2 . This line l_3 we want to construct must be the shared perpendicular bisector of AA' and BB' .



If we are familiar with conic sections, we will notice that the trace of the perpendicular bisector of AA' actually forms a parabola (A is the focus, l_1 is the directrix, the perpendicular bisector is the tangent of the parabola.). The same works for BB' . When we have these 2 parabolas, we must find the only perpendicular bisector (l_3) that is the shared tangent of both parabolas. Now the problem becomes clear: there is no way to construct this equation without using cubic. It is only a simple step in origami, but what's behind that is analytic geometry, cubic equation and many more serious mathematics. Now we have to admit: origami seems to be easy and childish, but it is more powerful than Euclid's ruler and compass.



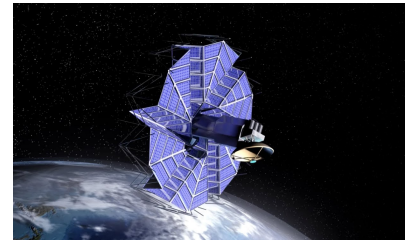
How can we apply such a powerful tool in real life? Mathematicians have made significant progress. One of the pioneers of origami mathematics is Robert J. Lang, an American mathematician and origami artist. In the late 1990s he created a software called TreeMaker, which was able to convert line drawing, or stick figure, into

a specific plan for a paper which can be folded in origami. With this powerful tool, mathematicians and artists can create super complicated and vivid model using a piece of uncut paper only. This is the technology that helps origami to really become a serious art form.



"Organist, Opus 363," by Robert J. Lang © Robert J. Lang

Origami has been applied to many more serious scientific area as well. The structures mathematicians has developed in origami turn out to have applications in medicine, in astronomy and many more. In building spaceships, scientists use some pattern in origami to design solar array so it can be folded down into a small package. In medicine, scientists have built something like “origami heart stent”, which can be folded using an origami pattern so it’s small enough to go through our blood vessels. The thing is, when scientists need something to be small enough to be carried or to go inside something, they will use some origami patterns.



Although origami has thousands of years of history, it was only associated with mathematics for merely about 100 years. But it was in this 100 years mathematicians have truly realized the power of origami and origami has helped scientists a lot in real world applications. Origami is just one example of mathematics - when you try to create something beautiful in maths for aesthetic value only, it is not useless in real world. Mathematics will reward you for doing something beautiful.

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